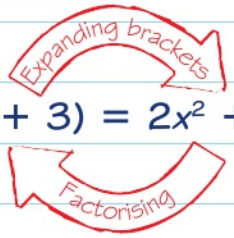
Factorising

Factorising is the opposite of expanding brackets.



You need to look for the largest factor you can take out of every term in the expression. This is what you need to write outside the brackets.

You have to factorise expressions as much as possible.

$10a^2 + 5a = 5(2a^2 + a)$
This expression has only been **partly factorised**.

$10a^2 + 5a = 5a(2a + 1)$
This expression has been **completely factorised**.
The two **factors** are $5a$ and $(2a + 1)$

To factorise ...

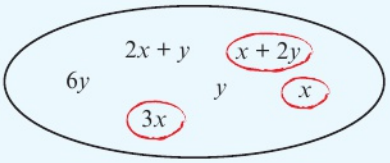
1. Look for the **largest factor** you can take out of every term.
   $12x - 8$
   The largest factor is 4.
2. Write this factor outside the brackets. Write the sign (+ or −) from the expression inside the brackets.
   $12x - 8 = 4(\quad - \quad)$
3. Work out what you need to multiply the factor by to get each term in the original expression.
   $12x - 8 = 4(3x - 2)$
4. **Check** your answer by expanding the brackets. You should get back to the original expression.
   $4 \times 3x = 12x$
   $4(3x - 2) = 12x - 8$ ✓
   $4 \times -2 = -8$

Examiners' report

When you write an expression with brackets both parts of the factorised expression are factors:
$6x^2 + 12xy = 6x(\underline{x + 2y}) = \underline{3x}(2x + 4y)$
$= \underline{x}(6x + 12y)$
So the three underlined expressions above are all factors of the original expression.

Real students have struggled with questions like this in recent exams – **be prepared!**

Worked example



Circle all the expressions above that are factors of $6x^2 + 12xy$ **(2 marks)**

Worked example

(a) Factorise $y^2 + 3y$ **(1 mark)**
$y^2 + 3y = y(y + 3)$

(b) Factorise fully $2p^2 - 4p$ **(2 marks)**
$2p^2 - 4p = 2p(p - 2)$

(a) **Check it!**
$y \times y = y^2$ ✓ $y \times 3 = 3y$ ✓
(b) The question says 'factorise fully'. This means you have to take the largest factor outside the brackets. $2p^2 - 4p = 2(p^2 - 2p)$ is not fully factorised.

Now try this

Worked solution video

1 Factorise
(a) $6n + 18$ **(1 mark)**
(b) $3p - 3$ **(1 mark)**

2 Factorise fully
(a) $4t^2 + 8t$ **(2 marks)**
(b) $3x^2 - 12x$ **(2 marks)**

3

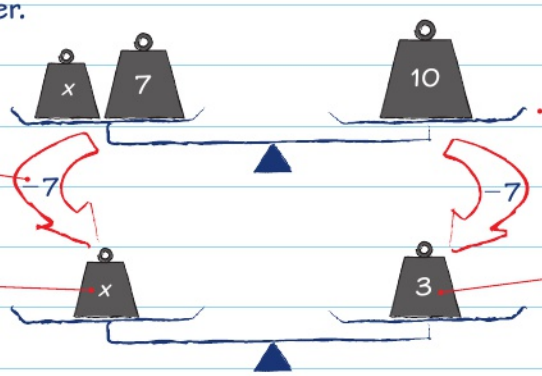
2	$4p$	$2p$	$p + 2q$	$p - 2q$

Write down the factors of $2p^2 - 4pq$ from the list. **(2 marks)**

Hot Topic

Linear equations 1

An **equation** is like a pair of scales. The equals sign tells you that the scales are **balanced**. The letter represents an unknown weight. You can solve the equation to find the value of the letter.



These scales represent the equation $x + 7 = 10$

You need to subtract 7 from both sides.

You need x on its own on one side.

The solution of this equation is $x = 3$

As long as you do the same thing to both sides, the scales stay balanced.

To **solve** an equation you need to get the letter on its own on one side.

It is important to write your working neatly when you are solving equations.

Every line of working should have an = sign in it.

Start a new line for each step. Do one operation at a time.

$5x + 3 = 18 \quad (-3)$

$5x = 15 \quad (\div 5)$

$x = 3$

Write down the operation you are carrying out. Remember to do the same thing to both sides of the equation.

$5x$ means $5 \times x$. You have to divide by 5 to get x on its own.

This is the solution to the equation. Your final line of working should look like this.

Worked example

(a) Solve $m + 6 = 15$ **(1 mark)**

$m + 6 = 15 \quad (-6)$

$m = 9$

(b) Solve $8p = 36$ **(1 mark)**

$8p = 36 \quad (\div 8)$

$p = 4.5$

In part (b) you need to do **two operations** to get a on its own. Do them one at a time, and make sure you show your working.

Worked example

(a) Solve $\frac{x}{3} = -5$ **(1 mark)**

$\frac{x}{3} = -5 \quad (\times 3)$

$x = -15$

(b) Solve $2a - 7 = 11$ **(2 marks)**

$2a - 7 = 11 \quad (+7)$

$2a = 18 \quad (\div 2)$

$a = 9$

Now try this

1 Solve

(a) $3y = 24$ **(1 mark)**

(b) $m + 8 = 25$ **(1 mark)**

(c) $31 - k = 12$ **(1 mark)**

2 Solve

(a) $4n - 7 = 13$ **(2 marks)**

(b) $2w + 11 = 32$ **(2 marks)**

3 (a) Solve $\frac{n}{4} = 9$ **(1 mark)**

(b) Solve $\frac{q}{11} = 7$ **(1 mark)**

Your answer doesn't have to be a whole number.

Worked solution video

Tricky Topic

Linear equations 2

Equations with brackets

Always start by multiplying out the brackets then collecting like terms.

For a reminder about multiplying out brackets have a look at page 28.

$2 \times 3y = 6y$

$2(3y + 5)$

$2 \times 5 = 10$

$$\begin{aligned} 2(3y + 5) &= 22 \\ 6y + 10 &= 22 \quad (-10) \\ 6y &= 12 \quad (\div 6) \\ y &= 2 \end{aligned}$$

Do one operation at a time. Write down the operation you are using at each step.

Equations with the letter on both sides

To solve an equation you have to get the letter on its own on one side of the equation.

Start by collecting like terms so that all the letters are together.

You can add or subtract multiples of x on both sides of the equation.

$$\begin{aligned} 4x + 26 &= 2 - 2x \quad (+2x) \\ 6x + 26 &= 2 \quad (-26) \\ 6x &= -24 \quad (\div 6) \\ x &= -4 \end{aligned}$$

Remember to do the same thing to both sides of the equation.

Worked example

Solve $7p + 2 = 5(p - 4)$ **(3 marks)**

$7p + 2 = 5(p - 4)$ (expand brackets)
$7p + 2 = 5p - 20$ $(-5p)$
$2p + 2 = -20$ (-2)
$2p = -22$ $(\div 2)$
$p = -11$

Start by expanding the brackets on the right-hand side. Then subtract $5p$ from both sides to get all the p terms together.

Examiners' report

Don't use a trial method to solve an equation. You probably won't find the correct answer, and you can't get any method marks.

Real students have struggled with questions like this in recent exams – **be prepared!**

Worked example

Solve $\dfrac{10 - y}{5} = 3$ **(3 marks)**

$\dfrac{10 - y}{5} = 3$ $(\times 5)$
$10 - y = 15$ $(+y)$
$10 = 15 + y$ (-15)
$-5 = y$

Start by multiplying both sides by 5 to remove the fraction. You want the y term to be positive so add y to both sides of the equation.

You can write your answer in the form $-5 = y$. As long as the letter is on its own on one side then the equation is solved.

Now try this

1 (a) Solve $33 - 5x = 8$ **(2 marks)**

(b) Solve $4 - 6x = 5 - 8x$ **(3 marks)**

Worked solution video

A positive x term is easier to work with. Start by adding $5x$ to both sides.

2 (a) Solve $3t - 7 = 5t + 15$ **(3 marks)**

(b) Solve $2(4h + 3) = 2$ **(3 marks)**

3 (a) Solve $3y + 19 = 5(y - 2)$ **(3 marks)**

(b) Solve $\dfrac{16 - 5m}{3} = 7$ **(3 marks)**

Expand the brackets then group the y terms on one side.